

Modal concepts in Gödel's metamathematics

Gödel-meeting, Bamberg Dec '25

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GÖDEL ENIGMA

Rediscovering Kurt Gödel through his unpublished works



The Gödel Enigma: Unveiling a Hidden Logical Heritage

Kurt Gödel's incompleteness theorems of 1931 are among the most iconic scientific achievements of the 20th century. Gödel's results led to the development of formal languages and algorithmic computability, upon which the first programming languages and computers were built two decades later. There are still several thousand pages of notes of this remarkable figure in the history of logic, left almost completely untouched, as they were written in an obsolete German stenographic script called Gabelsberger. The central aim of our project is to study these unpublished materials and make them available to future generations of logicians and philosophers. The ERC-funded project runs from 2018 to 2023.

Gödel's trial lecture, February 1933

Gödel published his well-known translation from intuitionistic logic to the modal logic **S4** in a laconic one-page article of 1933:

*If A is a provable formula of intuitionistic propositional logic, its modal translation $T(A)$ is provable in **S4**.*

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Über den intuitionistischen Aussagenkalkül

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Über den intuitionistischen Aussagenkalkül

Explains the details of his 1933 proof

Conjectures the converse:

*If the modal translation $T(A)$ of a formula A of intuitionistic propositional logic is provable in **S4**, A is provable in intuitionistic propositional logic.*

[illegible]

Brouwer has, as is known, put up the claim that the tertium is not a reliable means of proof in mathematics, with the motivation that it is, for many mathematical propositions, quite out of the question to decide today whether they are true or false, and maybe such a decision is on the whole impossible. The claim, however, that each proposition must be true or false, regardless of whether one can establish the one or the other, Brouwer takes to be senseless. Brouwer has further drawn various consequences from his claim, in a more or less systematic way. It is only his pupil Heyting who has taken this question to grips in an axiomatic way, namely by putting up an axiom system for logic, and more precisely, for the narrowest part of logic or the propositional calculus, in which the law of excluded middle cannot be derived.

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Kurt Gödel · Hans Hahn

Mathematical Logic in Vienna

Kurt Gödel (1906–1978) created a sensation by his incompleteness theorems of 1931, now seen as one of the high points of 20th century mathematics and science more generally. His professor at the University of Vienna Hans Hahn decided to organize a seminar on mathematical logic for the academic year 1931/32, with Gödel in charge for most of the practical side. The seminar proceedings, given in English translation in this book, are a unique witness of the state of research in logic and foundations of mathematics right after Gödel's theorems. They also add an important aspect to the intellectual history of these times in Vienna, both Hahn and Gödel having been members of the Vienna Circle. One fourth of the seminars were dedicated to Gödel's results: First the doctoral thesis about the completeness of predicate logic, then incompleteness, and last his results on intuitionistic logic. The seminars explain all these results in detail, in contrast to Gödel's publications of the time that often were quite laconic and extremely short. This book also contains Gödel's trial lecture on intuitionistic logic held in Vienna in 1933. The manuscript, recently found among the Gödel papers kept in Princeton, is preserved in Gödel's forgotten German shorthand and published here in an English translation.

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Gödel · Hahn

Mathematical Logic in Vienna



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Kurt Gödel
Hans Hahn

Mathematical Logic in Vienna

Edited by
Jan von Plato



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Gödel's 1933 translation

$p \vee q$	$B(p) \vee B(q)$	} nichts gew. Bzgl.
$p \supset q$	$B(p) \supset B(q)$	
$p \wedge q$	$B(p) \wedge B(q)$	
$\neg p$	$\neg B(p)$	

$$p \vee \neg p \quad B(p) \vee B(\neg B(p))$$

p entw. bew. oder nachweislich
unbew.

$$\neg \neg p \quad \neg \neg B(p)$$

p nicht nachweislich unbeweisbar

Proof of the conjecture in Resultate Grundlagen, 1941

240

2) $\sim A \vdash A: \text{Lp}$

$$M[A_1 \dots A_n]' \equiv_H M'[A_1' \dots A_n']$$

$$< M' = \text{Hyp. Ident.} \leq M - \text{neg.}$$

(U Th 62)

13 Hyp 11, 12, 13 $\vdash A, B \in \text{Loris}$

$$(\mathcal{L}(A) \supset \mathcal{L}(A))' \sim \text{Hyp. 1}$$

$$(\mathcal{L}(A), \mathcal{L}(A \supset B) \supset \mathcal{L}(B))' \sim \text{Hyp. 2}$$

$$\text{Bar. 1. } \vdash \vdash_H \mathcal{L}(A) \supset \mathcal{L}(A) = A \supset A'$$

$$2 \vdash \vdash_H A' (A \supset B)' \supset B' \text{ ab } \checkmark \text{ U Hyp 10}$$

14 Hyp 11 $\vdash A' \supset (A \supset B)' \text{ Hyp. 1. } \supset B'$

U. Hyp 10

15 Hyp 11 $\vdash A' \supset \text{Hyp. 1. } \supset \mathcal{L}(A)'$

$$\vdash \mathcal{L}(A)' = A'$$

Th 63

$$\vdash A \supset \text{Loris} \text{ Hyp. 11, 12, 13}$$

Bar.

$$\text{U Hyp 11, 12, 13 } \vdash \mathcal{L}(A) \supset \text{Loris}$$

$$\text{U Hyp 14, 15 } \vdash \text{Loris}$$

Th 64

$$\vdash A \supset \text{Hyp. 11, 12, 13 } \supset V(A)$$

$$[\text{vgl. Hyp 9}] \supset \text{Loris} \text{ Hyp. 11, 12, 13 } \supset A \supset \text{Hyp. 11}$$

Bar.

$$\text{U Th 63 } \supset V(A)' \supset \text{Hyp. 11, 12, 13}$$

$$< V(A)' \equiv_H A \supset \text{Hyp. 9.}^*$$

$\mathcal{L}_1 \sim \text{Modalitäten im Lorisstruktur}$

Schluss an $\sim \supset \text{Loris}$ in Modalitäten

U. in m' Implik. $\supset$

$\supset \text{Loris}$ in Modalitäten



* Forts. p. 249.

Sources and Studies in the History of Mathematics and Physical Sciences

Maria Hämeen-Anttila · Jan von Plato Editors

Kurt Gödel: Results on Foundations

Kurt Gödel (1906–1978) gained world-wide fame by his incompleteness theorem of 1931. Later, he set as his aim to solve what are known as Hilbert's first and second problems, namely Cantor's continuum hypothesis about the cardinality of real numbers, and secondly the consistency of the theory of real numbers and functions. By 1940, he was halfway through the first problem, in what was his last published result in logic and foundations. His intense attempts thereafter at solving these two problems have remained behind the veil of a forgotten German shorthand he used in all of his writing.

Results on Foundations is a set of four shorthand notebooks written in 1940–42 that collect results Gödel considered finished. Its main topic is set theory in which Gödel anticipated several decades of development. Secondly, Gödel completed his 1933 program of establishing the connections between intuitionistic and modal logic, by methods and results that today are at the same time new and 80 years old.

The present edition of Gödel's four notebooks encompasses the 368 numbered pages and 126 numbered theorems of the *Results on Foundations*, together with a list of 74 problems on set theory Gödel prepared in 1946, and a list of an unknown date titled "The grand program of my research in ca. hundred questions."

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Kurt Gödel: Results on Foundations

Sources and Studies in the History of Mathematics
and Physical Sciences

Maria Hämeen-Anttila
Jan von Plato Editors

Kurt Gödel: Results on Foundations

 Springer

Table 1. Gödel's provability axioms.

$$1. \Box A \supset A$$

$$2. \Box(A \supset B) \supset (\Box A \supset \Box B)$$

$$3. \Box A \supset \Box \Box A$$

Rule of inference: If A has been proved, $\Box A$ can be inferred.

Table 2. Natural modal rules, calculus S4.

$$\frac{\begin{array}{c} \Box \Gamma \\ \vdots \\ A \end{array}}{\Box A} \Box I$$

$$\frac{\begin{array}{c} 1 \\ A \\ \vdots \\ C \end{array}}{\Box A} \Box E, 1$$

Modal translation

Table 3. The modal translation.

1. $T(P) = \Box P$
2. $T(A \& B) = \Box(T(A)) \& \Box(T(B))$
3. $T(A \vee B) = \Box(T(A)) \vee \Box(T(B))$
4. $T(A \supset B) = \Box(T(A)) \supset \Box(T(B))$
5. $T(\neg A) = \neg \Box(T(A))$

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Example:

$$T((A \supset \neg B) \supset (B \supset \neg A)) = \Box(\Box A \supset \Box \neg \Box B) \supset \Box(\Box B \supset \Box \neg \Box A)$$

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S-formula: main connective $\Box$

I-formula: main connective $\&, \vee, \supset, \neg$

Note the alternating **I-S-I...** subformula structure

Classical natural deduction with primitive negation

Treat negation as a primitive notion, with the rules:

$$\frac{\begin{array}{c} 1 \\ A \\ \vdots \\ C \end{array} \quad \begin{array}{c} 1 \\ A \\ \vdots \\ \neg C \end{array}}{\neg A} \neg I, 1 \qquad \frac{\begin{array}{c} 1 \\ \neg A \\ \vdots \\ C \end{array} \quad \begin{array}{c} 1 \\ \neg A \\ \vdots \\ \neg C \end{array}}{A} \neg E, 1$$

Table 4. The rules for negation, calculus NK.

Classical natural deduction with primitive negation

Treat negation as a primitive notion, with the rules:

$$\begin{array}{c} \begin{array}{c} 1 \quad 1 \\ A \quad A \\ \vdots \quad \vdots \\ C \quad \neg C \end{array} \quad \neg I, 1 \\ \hline \neg A \end{array} \qquad \begin{array}{c} \begin{array}{c} 1 \quad 1 \\ \neg A \quad \neg A \\ \vdots \quad \vdots \\ C \quad \neg C \end{array} \quad \neg E, 1 \\ \hline A \end{array}$$

Table 4. The rules for negation, calculus NK.

With negation defined by $\neg A = A \supset \perp$, the standard rule of indirect proof and *ex falso* are the special cases:

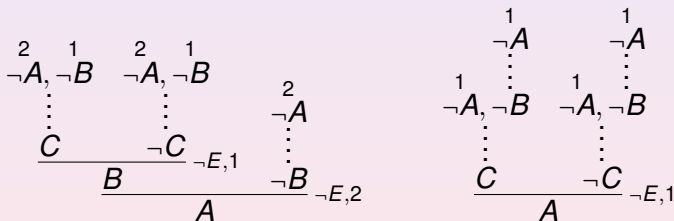
$$\begin{array}{c} \begin{array}{c} 1 \quad 1 \\ \neg A \quad \neg A \\ \vdots \quad \vdots \\ \perp \quad \neg \perp \end{array} \quad \neg E, 1 \\ \hline A \end{array} \qquad \begin{array}{c} \begin{array}{c} \vdots \quad \vdots \\ C \quad \neg C \end{array} \quad \neg E \\ \hline A \end{array}$$

In the first, the second premiss $\neg \perp$ is a theorem, so delete it.
The second shows that **NI** is special case of **NK**

Theorem. Normalization for NK. *Derivations in NK convert to normal form.*

Proof. The cases to consider, beyond known proofs of normalization for **NI** concern rules $\neg I$ and $\neg E$:

1. Permutative conversion, with the left premiss of $\neg E$ derived by $\neg E$. The derivation and its transformation are:



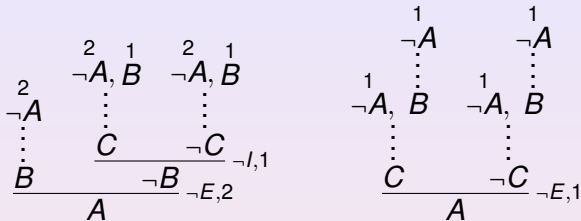
Successive steps of indirect proof collapse into just one

2. Detour conversion, with the right premiss of $\neg E$ derived by $\neg I$. The derivation and its transformation are:

$$\begin{array}{c}
 \begin{array}{c} 2 \\ \vdots \\ \neg A \end{array} \quad \begin{array}{c} 2 \quad 1 \\ \vdots \\ \neg A, B \\ \vdots \\ C \end{array} \quad \begin{array}{c} 2 \quad 1 \\ \vdots \\ \neg A, B \\ \vdots \\ \neg C \end{array} \\
 \hline
 \begin{array}{c} B \quad \neg B \\ \neg E, 2 \end{array} \quad \neg I, 1 \\
 \hline
 A
 \end{array}$$

$$\begin{array}{c}
 \begin{array}{c} 1 \\ \vdots \\ \neg A \end{array} \quad \begin{array}{c} 1 \\ \vdots \\ \neg A, B \\ \vdots \\ C \end{array} \quad \begin{array}{c} 1 \\ \vdots \\ \neg A \end{array} \quad \begin{array}{c} 1 \\ \vdots \\ \neg A, B \\ \vdots \\ \neg C \end{array} \\
 \hline
 \begin{array}{c} C \quad \neg C \\ \neg E, 1 \end{array} \\
 \hline
 A
 \end{array}$$

2. Detour conversion, with the right premiss of $\neg E$ derived by $\neg I$. The derivation and its transformation are:



Right premiss of $\neg E$ not derived by $\neg I$

2. Detour conversion, with the right premiss of $\neg E$ derived by $\neg I$. The derivation and its transformation are:

Diagram illustrating the derivation of $\neg A$ from A using the rules of propositional logic. The diagram shows two separate derivations, each starting with A and leading to $\neg A$.

Left Derivation:

- Start with A (bottom line).
- Assume $\neg B$ (second line from bottom).
- Derive C (third line from bottom).
- Derive $\neg C$ (fourth line from bottom).
- Derive $\neg A, B$ (fifth line from bottom).
- Derive $\neg A$ (top line).

Right Derivation:

- Start with A (bottom line).
- Assume $\neg C$ (second line from bottom).
- Derive C (third line from bottom).
- Derive $\neg A, B$ (fourth line from bottom).
- Derive $\neg A$ (top line).

Right premiss of $\neg E$ not derived by $\neg I$

3. Major premiss of $\&E$, $\vee E$, or $\supset E$ derived by $\neg E$: The conversions are similar to those for the rule of indirect proof in von Plato and Siders (2012). QED.

With **NKS4**, normal derivability requires:

Normality. *The major premiss of $\Box E$ must be an assumption.*

Theorem. Normalization for NKS4. *Derivations in NKS4 convert to normal form.*

Proof. The only new case to consider is when the major premiss of $\Box E$ has been derived by $\neg E$. The derivation is:

$$\begin{array}{c}
 \begin{array}{c} 1 \\ \neg \Box A \\ \vdots \\ B \end{array} \quad \begin{array}{c} 1 \\ \neg \Box A \\ \vdots \\ \neg B \end{array} \\
 \hline
 \Box A \quad \neg E,1
 \end{array}
 \quad \begin{array}{c} 2 \\ A \\ \vdots \\ C \end{array}$$

$$\frac{\Box A \quad C}{C} \Box E,2$$

Theorem. Normalization for NKS4. *Derivations in NKS4 convert to normal form.*

Proof. The only new case to consider is when the major premiss of $\Box E$ has been derived by $\neg E$. The derivation is:

$$\begin{array}{c}
 \begin{array}{c} 1 \\ \vdots \\ \neg \Box A \\ B \end{array} \quad \begin{array}{c} 1 \\ \vdots \\ \neg \Box A \\ \neg B \end{array} \quad \begin{array}{c} 2 \\ \vdots \\ A \\ C \end{array} \\
 \hline
 \Box A \quad \neg E,1 \\
 \hline
 C \quad \Box E,2
 \end{array}$$

The transformation is into:

$$\begin{array}{c}
 \begin{array}{c} 1 \\ \vdots \\ A \\ C \end{array} \quad \begin{array}{c} 2 \\ \vdots \\ \Box A \\ C \end{array} \quad \begin{array}{c} 5 \\ \vdots \\ \neg C \end{array} \\
 \hline
 \neg \Box A \quad \neg I,2 \\
 \vdots \\
 B
 \end{array} \quad \begin{array}{c}
 \begin{array}{c} 3 \\ \vdots \\ A \\ C \end{array} \quad \begin{array}{c} 4 \\ \vdots \\ \Box A \\ C \end{array} \quad \begin{array}{c} 5 \\ \vdots \\ \neg C \end{array} \\
 \hline
 \neg \Box A \quad \neg I,4 \\
 \vdots \\
 \neg B \quad \neg E,5
 \end{array} \\
 \hline
 C$$

Corollary. CM-lemma. *If in a normal derivation of a theorem in **NKS4** the last rule is $\neg E$, the derivation of its right premiss is degenerate.*

By the degeneracy, $C = A$, and with the right premiss left unwritten, we have:

$$\frac{\begin{array}{c} 1 \\ \neg A \\ \vdots \\ C \end{array} \quad \begin{array}{c} 1 \\ \neg A \\ \vdots \\ \neg C \end{array}}{A} \neg E,1$$

$$\frac{\begin{array}{c} 1 \\ \neg A \\ \vdots \\ A \end{array} \quad \begin{array}{c} 1 \\ \neg A \end{array}}{A} \neg E,1$$

$$\frac{\begin{array}{c} 1 \\ \neg A \\ \vdots \\ A \end{array}}{A} CM,1$$

Here *CM* stands for *Consequentia Mirabilis*, a rule that corresponds to the axiom $(\neg A \supset A) \supset A$. The CM-lemma holds in particular for **NK**.

Example: law of excluded middle

$$\frac{\frac{\frac{A^1}{A \vee \neg A} \vee I}{\frac{\neg A}{A \vee \neg A} \vee I} \neg(A \vee \neg A)^2 \neg I,1}{A \vee \neg A} \neg E,2$$

Main Lemma. Conservativity of NKS4 over NIS4 for T-formulas. *If the translation $T(C)$ of an intuitionistic formula C is derivable in **NKS4**, it is already derivable in **NIS4**.*

Proof. Consider an uppermost step of $\neg E$ in a normal derivation in **NKS4**:

$$\frac{\begin{array}{c} 1 \\ \neg A \\ \vdots \\ B \end{array} \quad \begin{array}{c} 1 \\ \neg A \\ \vdots \\ \neg B \end{array}}{A} \neg E, 1$$

In the derivation of the right premiss, $\neg B = \neg A$:

Assume the assumption $\neg A$ is not vacuous.

$\neg A$ not a premiss in $\Box I$, $\&I$, $\vee I$, $\supset I$:

- $\Box I$ excluded by its restriction
- other I -rules violate the **S-I** subformula structure

It follows that $\neg A$ is a premiss in rule $\neg I$.

A not a negation, therefore $\neg B = \neg A$, so $B = A$ and we have:

$$\begin{array}{c}
 1 \\
 \neg A \\
 \vdots \\
 A \quad \begin{array}{c} 1 \\ \neg A \end{array} \\
 \hline
 A \quad \neg E, 1
 \end{array}$$

If A is the conclusion of the whole derivation, it is an I-formula, and the same if it is a premiss in rule $\Box I$. Then, if A is a negation, the step of indirect inference is not needed. Therefore A is equal to one of $\Box D \& \Box E$, $\Box D \vee \Box E$, or $\Box D \supset \Box E$, and A is derived by $\& I$, $\vee I$, or $\supset I$. In each of these rules, the premisses contain one of $\Box D$, $\Box E$, but in the presence of the negative assumption $\neg A$, these cannot have been derived by rule $\Box I$. Therefore the assumption $\neg A$ is not used in the derivation of the premiss A and the step of $\neg E$ can be left out to obtain a derivation of A in **NIS4**.

Finally, in the postponed case that $\neg A$ is absent in the derivation of the right premiss of $\neg E$, B must have the form $\Box D$. Its derivation in the left premiss of $\neg E$ must be by rule $\Box I$ and that can be, just as before, only if the assumption $\neg A$ was absent in the derivation of the left premiss. QED.

Finally, in the postponed case that $\neg A$ is absent in the derivation of the right premiss of $\neg E$, B must have the form $\Box D$. Its derivation in the left premiss of $\neg E$ must be by rule $\Box I$ and that can be, just as before, only if the assumption $\neg A$ was absent in the derivation of the left premiss. QED.

The alternating **S-I** structure of the subformulas imposes what I like to call “Gödel’s modal straitjacket.”

The converse translation

Example:

$$T((A \supset \neg B) \supset (B \supset \neg A)) = \Box(\Box A \supset \Box \neg \Box B) \supset \Box(\Box B \supset \Box \neg \Box A)$$

Start from uppermost formulas, delete all the modal operators and instances of the modal rules, and write just those of **NI**.

Step 1:

[illegible]

Step 2:

$$\frac{\frac{\frac{5}{B} \quad \frac{\frac{A \supset \neg B}{3} \quad \frac{A \quad \neg B}{4}}{\neg B} \supset E, 2}{\neg \Box A} \neg I, 4}{\Box \neg \Box A} \Box I}{\Box B \supset \Box \neg \Box A} \supset I, 5}{\Box (\Box B \supset \Box \neg \Box A)} \Box I}{\Box (\Box A \supset \Box \neg \Box B) \supset \Box (\Box B \supset \Box \neg \Box A)} \supset I, 6$$

Step 3:

$$\frac{\frac{\frac{5}{B} \quad \frac{\frac{3}{A \supset \neg B} \quad \frac{4}{A} \quad \frac{2}{\neg B}}{\supset E, 2}}{\neg B}}{\neg A} \neg I, 4$$

$$\frac{\frac{\frac{\frac{\Box B \supset \Box \neg \Box A}{\supset I, 5}}{\Box(\Box B \supset \Box \neg \Box A)} \Box I}{\Box(\Box A \supset \Box \neg \Box B) \supset \Box(\Box B \supset \Box \neg \Box A)} \supset I, 6$$

Step 4:

$$\frac{\frac{\frac{A \supset \neg B \quad A \quad \neg B}{\supset E, 2}}{B \quad \neg B}{\neg I, 4}}{\neg A}{\supset I, 5}}{(A \supset \neg B) \supset (B \supset \neg A)}{\supset I, 6}$$

[illegible]

$$\frac{\frac{\frac{A \supset \neg B \quad A \quad \neg B}{\supset E, 2}}{B \quad \neg B}{\neg I, 4}}{\neg A}{\supset I, 5}}{\frac{B \supset \neg A}{(A \supset \neg B) \supset (B \supset \neg A)}}{\supset I, 6}$$

And next:

Modal concepts in Gödel's *Arbeitshefte* 4–16 on logic and foundations, 1940–42

Gödel was greatly impressed by the following fact:

The opens of a topological space form a Heyting algebra

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Gödel's modal interpretation of intuitionistic logic gave a bridge between provability logic and topological models.

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Gödel was greatly impressed by the following fact:

The opens of a topological space form a Heyting algebra

Gödel's modal interpretation of intuitionistic logic gave a bridge between provability logic and topological models.

Gödel found a result by Steinhaus of 1923 that in logical terms states:

There is an infinity of intuitionistically inequivalent formulas with one formula variable.

The result is otherwise known from Rieger (1949) and Nishimura (1960).

Gödel-Löb logic?

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Löb's axiom: $\Box(\Box A \supset A) \supset \Box A$

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$\Box A \supset A$ replaced by Löb's axiom gives the provability logic **GL**.

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Martin Löb in 1955 gave a logic of provability that does not assume consistency:

Löb's axiom: $\Box(\Box A \supset A) \supset \Box A$

$\Box A \supset A$ replaced by Löb's axiom gives the provability logic **GL**.

Open question: To what extent is **GL** anticipated in the *AH*?

Heute wir zeigen $\varphi(x) \leftrightarrow \varphi(y) \vdash \exists y (x \in M \wedge \varphi(y))$
 $x \sim y \supset \varphi(x) \equiv \varphi(y)$

Ann. $B \sim B \supset P$ $B \sim P \supset \sim P$ Ann. es ist $A \vdash \varphi$
 $P \supset \sim B \supset P$ $B \supset P$

$P \wedge A \vdash B \sim B$ by Ext. These

$B \sim B \wedge A \sim S P_{\varphi} \quad \varphi \text{ ist } A$

$\langle A \wedge \sim B \rangle \vdash \varphi$ by Ext. These

$A \vdash (M) \vdash C \vdash M \wedge A \wedge A \vdash C$

$(M) \vdash (A \wedge B) \vdash (A \wedge B \supset A \wedge M \equiv B \wedge M) \supset$

Neuer Axiom System f. alle B-Terme:

Th 1.1 $S_B : A \vdash B(A) \supset \varphi$ $\varphi \in \mathcal{L}$

2. $B(\varphi) [B(\varphi) \supset \varphi]$ $B(\varphi) [B(\varphi) \supset B(\varphi)]$

3. $B(\varphi) [B(\varphi) \supset \varphi]$ $B(\varphi) \supset B(\varphi)$

4. $B(\varphi) \varphi$ $\varphi \in \mathcal{L}^M \supset \varphi \in \mathcal{L}^M$

5. Implik. $\downarrow$ $\varphi \supset \psi$

1. $\varphi \in \mathcal{L}$
 2. $\varphi \in \mathcal{L}^M$

Bem 1 At. 2.1 $\forall a, \exists y \in B$ p. 2.2 a,
 e f a n 2.2 s. 1.0 (a, 2.2 a, 2.2 a
 2.2 a - m. 1.0 s. 1.0 a, 2.2 a, 2.2 a
 m. 1.0)

Bem 2 p. m. 1.0 a 1.5 2.1 m. 1.0

$B(B(a)) \supset B(a)$ 2.2 1.3. 1.0 $B(y)$

m. 1.0 a 1.5 2.1 m. 1.0

Wegleitung $B(a) \supset a$ 1.5 2.1 m. 1.0 $B(y)$

Bem 3. m. 1.5 2.1 m. 1.0 2.2, 3, 4 $B(y)$

p. Beh. a 1.5 2.1 m. 1.0 s. 1.0 a 1.5 2.1 m. 1.0

(1.5 2.1 m. 1.0 s. 1.0 a 1.5 2.1 m. 1.0)

a 1.5 2.1 m. 1.0 s. 1.0 a 1.5 2.1 m. 1.0

2.2 a 1.5 2.1 m. 1.0 s. 1.0 a 1.5 2.1 m. 1.0

1.5 2.1 m. 1.0 s. 1.0 a 1.5 2.1 m. 1.0

Bem 4. $\forall U \in \mathcal{L}_K = B^*(U) \in \mathcal{L}$ y 1.5 2.1 m. 1.0

* 1.5 2.1 m. 1.0 s. 1.0 a 1.5 2.1 m. 1.0

Gödel's use of modal logic in the *AH* to be analyzed by systems such as

NKS4, NIS4, NKGL, NIGL

Gödel uses modal logic for models of

-intuitionistic type theory in which AC fails

-Zermelo's set theory

Independence of the axiom of choice by the B-method⁶⁰

14/X.42

S_B

Definition. 1. Meaningful B-formula. Built up by means of:

- 1.) Variables for natural numbers and functions of arbitrary type of the Church-Lesniewskian type theory⁶¹
- 2.) f (= successor), 0.
- 3.) $=$ (for individuals).
- 4.) Quantifiers and operations of the propositional calculus.
- 5.) B (for propositions).⁶²
- 6.) $\neg$ ⁶³

Definition 1.' $S \subset S_B$ (without the sign B)

Definition 2. Usual axioms. (A subclass of S , $Ax \subseteq S_B$).

Logical axioms:

- 1.) All that result by substitution in formulas of the propositional calculus. Also for formulas that contain B !
- 2.) $(u)\varphi(u) \supset \varphi(a)$ for arbitrary variables u, a .⁶⁴ (a can be also the constant f).

Comprehension axiom:

- 3.) $(\exists f)B(x)[f(x) \equiv \mathfrak{A}(x)]$ if $\mathfrak{x}$ is a complete argument series for f .